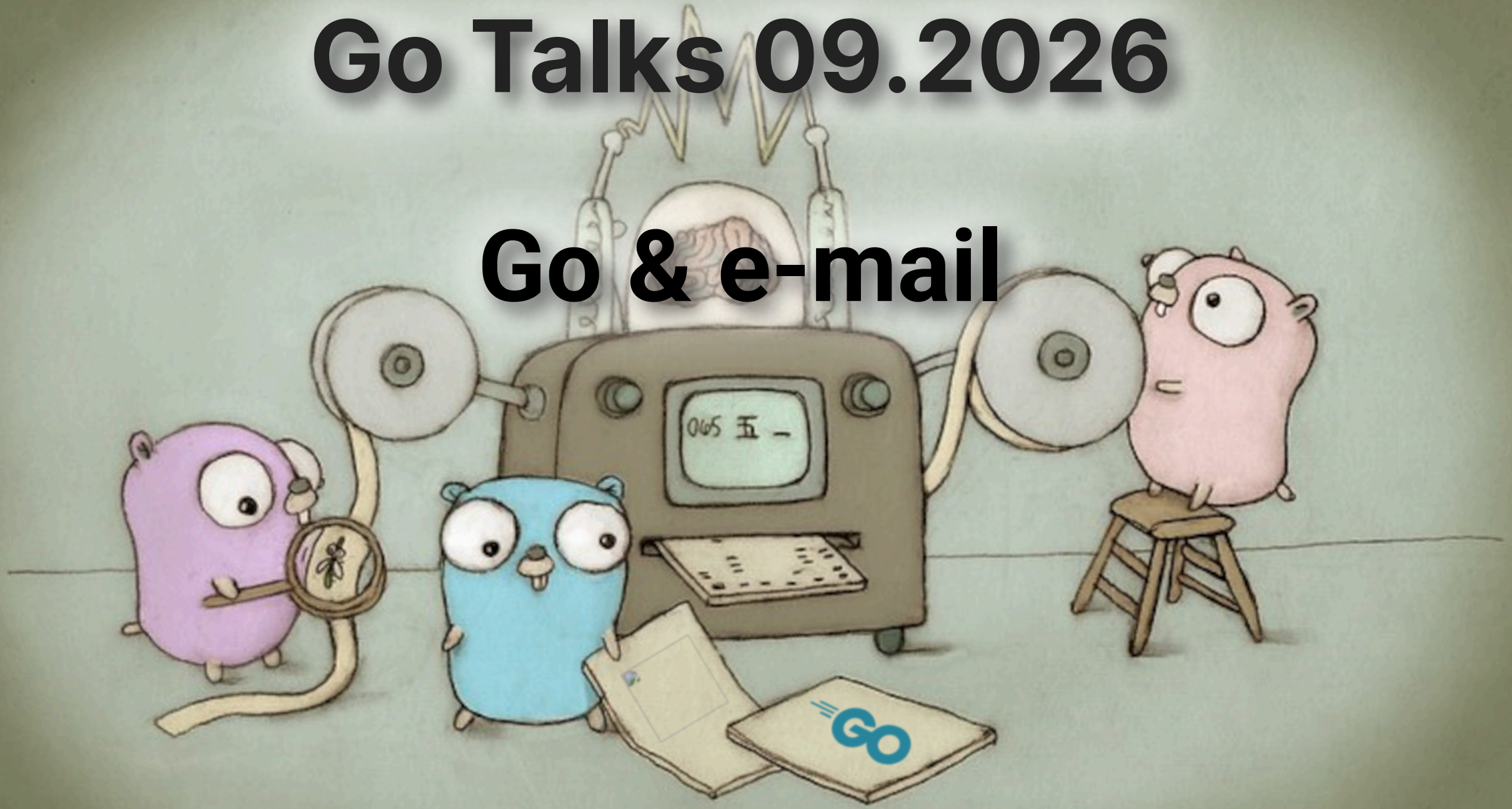
Go Talks 09.2026

Go & e-mail



How to reach us



meetup.com/Golang-ZG/



@golangzg



github.com/golang-zg/meetups



@golangzg.bsky.social



invite.slack.golangbridge.org #croatia

Go news

Go 1.27

go.dev/blog/go1.27



victoriametrics.com/blog/go-1-27/



Go news



Generic Methods

- Methods on concrete types can now declare their own type parameters.
- allows generic operations within type namespaces rather than requiring package-level functions.

- *Interface methods still cannot declare type parameters*

Go news

```go
type Cache[T any] struct {
    data map[string]T
}

func NewCache[T any]() *Cache[T] {
    return &Cache[T]{data: make(map[string]T)}
}

func (c *Cache[T]) Set(key string, val T) {
    c.data[key] = val
}

func (c *Cache[T]) Get(key string) (T, bool) {
    val, ok := c.data[key]
    return val, ok
}
```

```go
func main() {
    intCache := NewCache[int]()
    intCache.Set("port", 8080)

    val, ok := intCache.Get("port")
    if ok {
        fmt.Print(val)
    }
}
```

```
$ go run .
8080
```

Go news

Direct Embedded Field Initialization

- Keys in struct composite literals can now be valid field selectors, allowing direct initialization of fields inside embedded or nested structs without writing inner struct literals.

```go
type Address struct {
    City    string
}

type User struct {
    Address
    Name    string
}
```

Go 1.26

```go
u := User{
    Name: "Alice",
    Address: Address{
        City:    "New York",
    },
}
```

Go news

Direct Embedded Field Initialization

- Keys in struct composite literals can now be valid field selectors, allowing direct initialization of fields inside embedded or nested structs without writing inner struct literals.

```go
type Address struct {
    City    string
}

type User struct {
    Address
    Name    string
}
```

Go 1.27

```go
u := User{
    Name: "Alice",
    City:    "New York",
}
```

Go news

Faster memory allocation

- The compiler now generates calls to size-specialized memory allocation routines, cutting the cost of some small (under 80 bytes) allocations by up to 30%.
- Improvements vary with the workload, but the overall gain is expected to be around 1% in real allocation-heavy programs. The tradeoff is about 60 KB of extra binary size, independent of the workload.

Go & e-mail

github.com/orgs/golang-mail/repositories

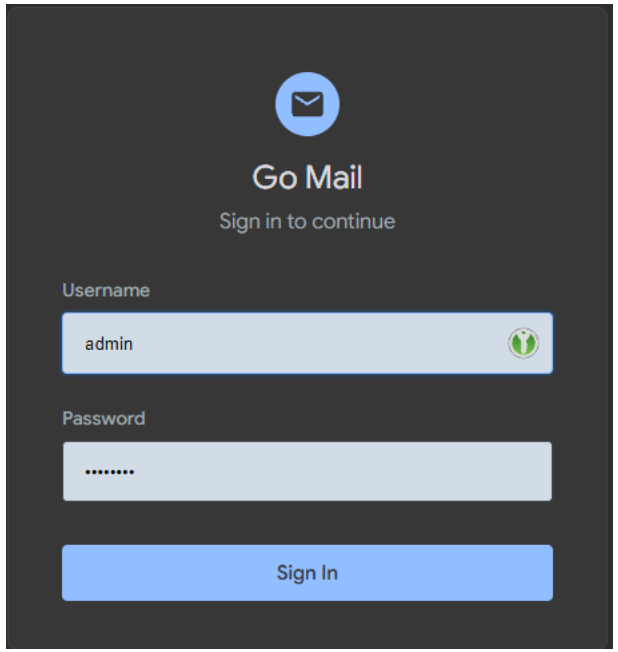
github.com/golang-mail/go-mail

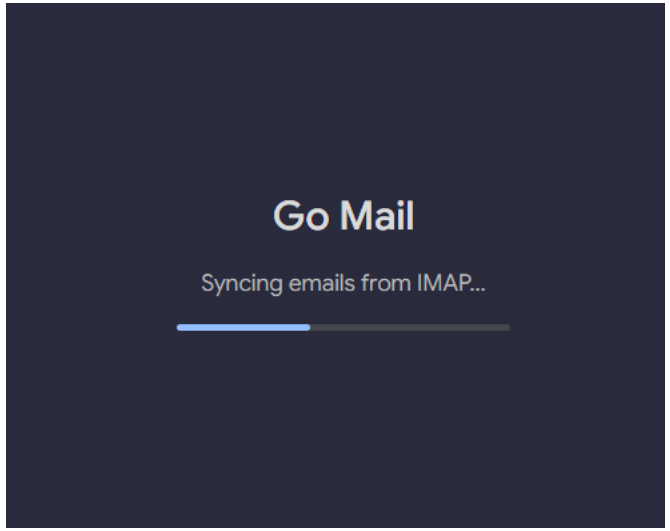


Go & e-mail

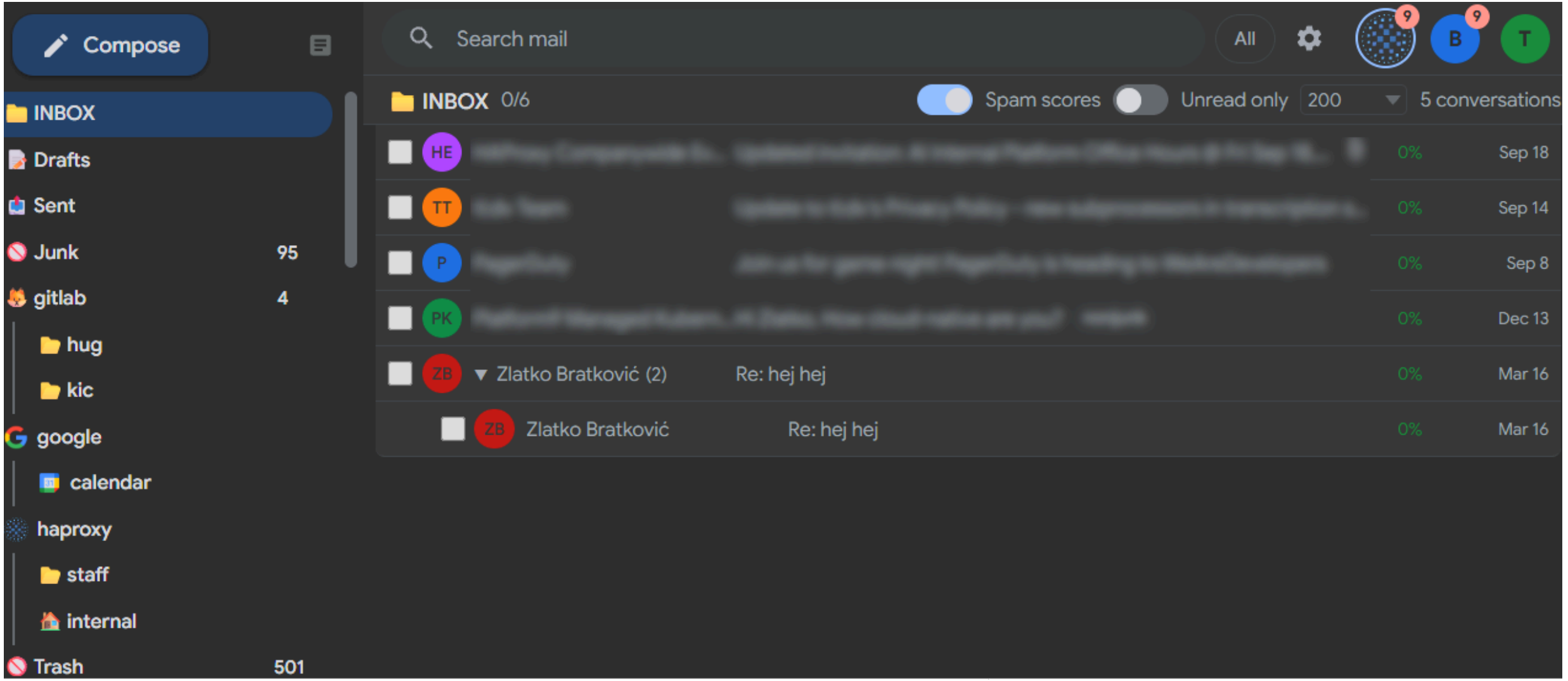
- written in go
- browser based UI - javascript
- compose, reply, reply all
- filters
- plugins
- spam detection
- multi accounts
- AI 'friendly'
  - multiple storage types

go-mail


Go Mail
Sign in to continue
Username
admin
Password
Sign In


Go Mail
Syncing emails from IMAP...

go-mail


Compose
Search mail
All
INBOX 0/6
Spam scores
Unread only
200
5 conversations
INBOX
Drafts
Sent
Junk 95
gitlab 4
hug
kic
google
calendar
haproxy
staff
internal
Trash 501
HE
TT
P
PK
ZB
Zlatko Bratković (2)
Re: hej hej
Zlatko Bratković
Re: hej hej
0%
Sep 18
Sep 14
Sep 8
Dec 13
Mar 16
Mar 16

go-mail

67% spam

[SPAM] Oral-B for a Brighter Smile - Gift from UnitedHealthcare

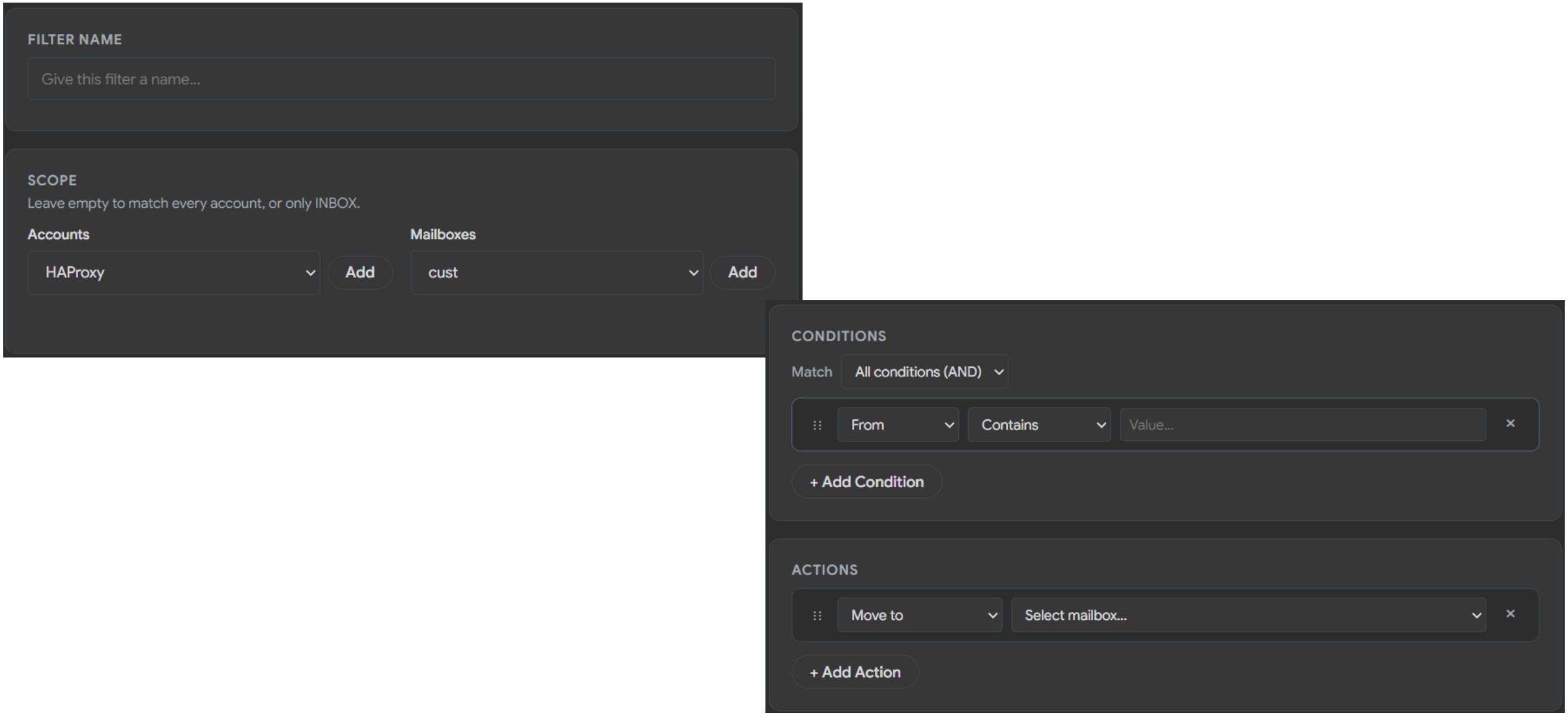
UnitedHealthcare Oral Care Gift <UnitedHealthcareOralCareGift@fatalblackoutgear.shop> Sat, Sep 19, 2026, 08:15 AM
to zbratkovic@haproxy.com
reply-to: Claim Your Oral-B Dental Kit (UnitedHealthcareOralCareGift@fatalblackoutgear.shop)

Not seeing the email as expected? View it online.

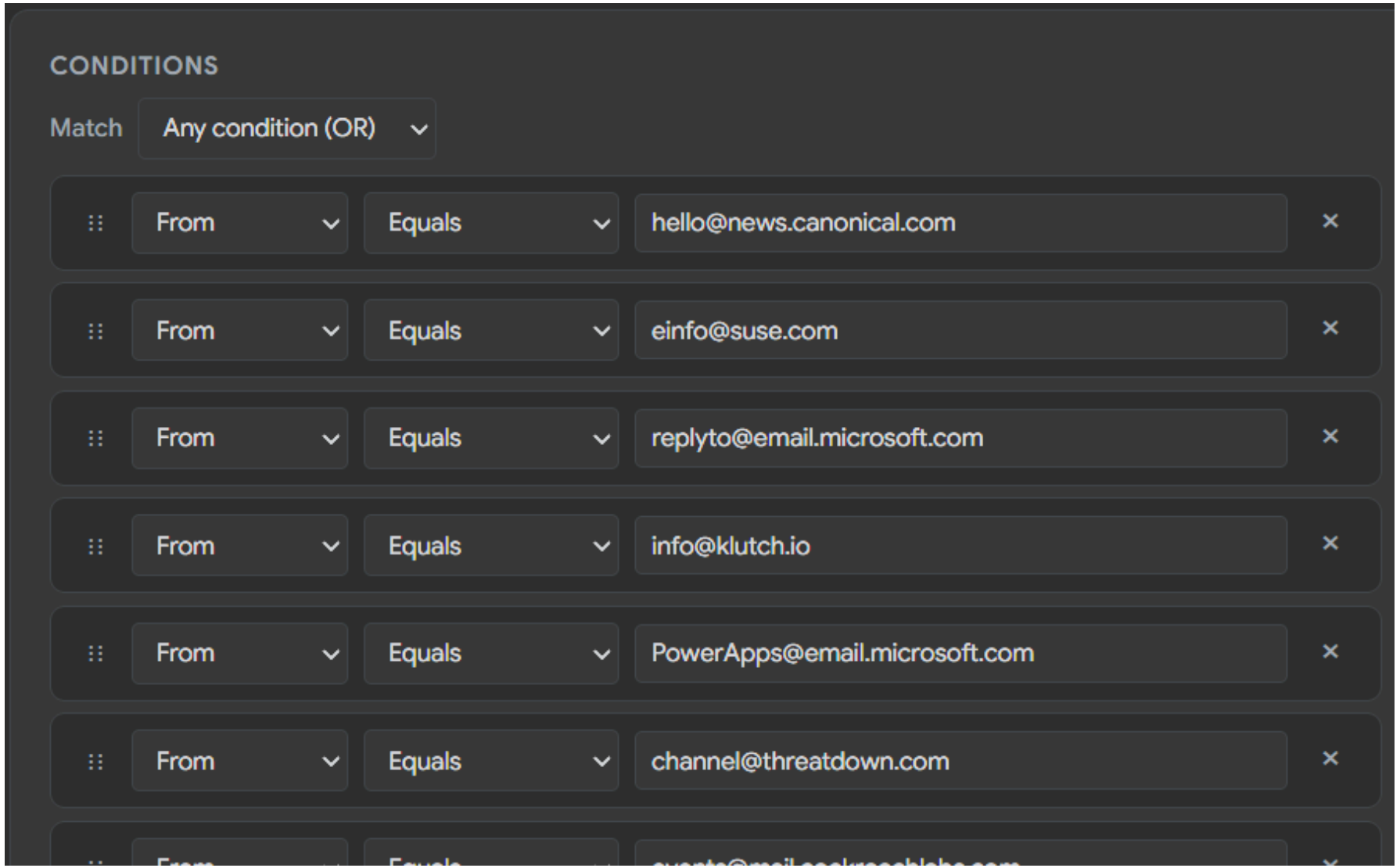
Oral-B for a Brighter Smile - Gift from UnitedHealthcare

United Healthcare
Oral-B Dental Kit

go-mail - filters



go-mail - filters

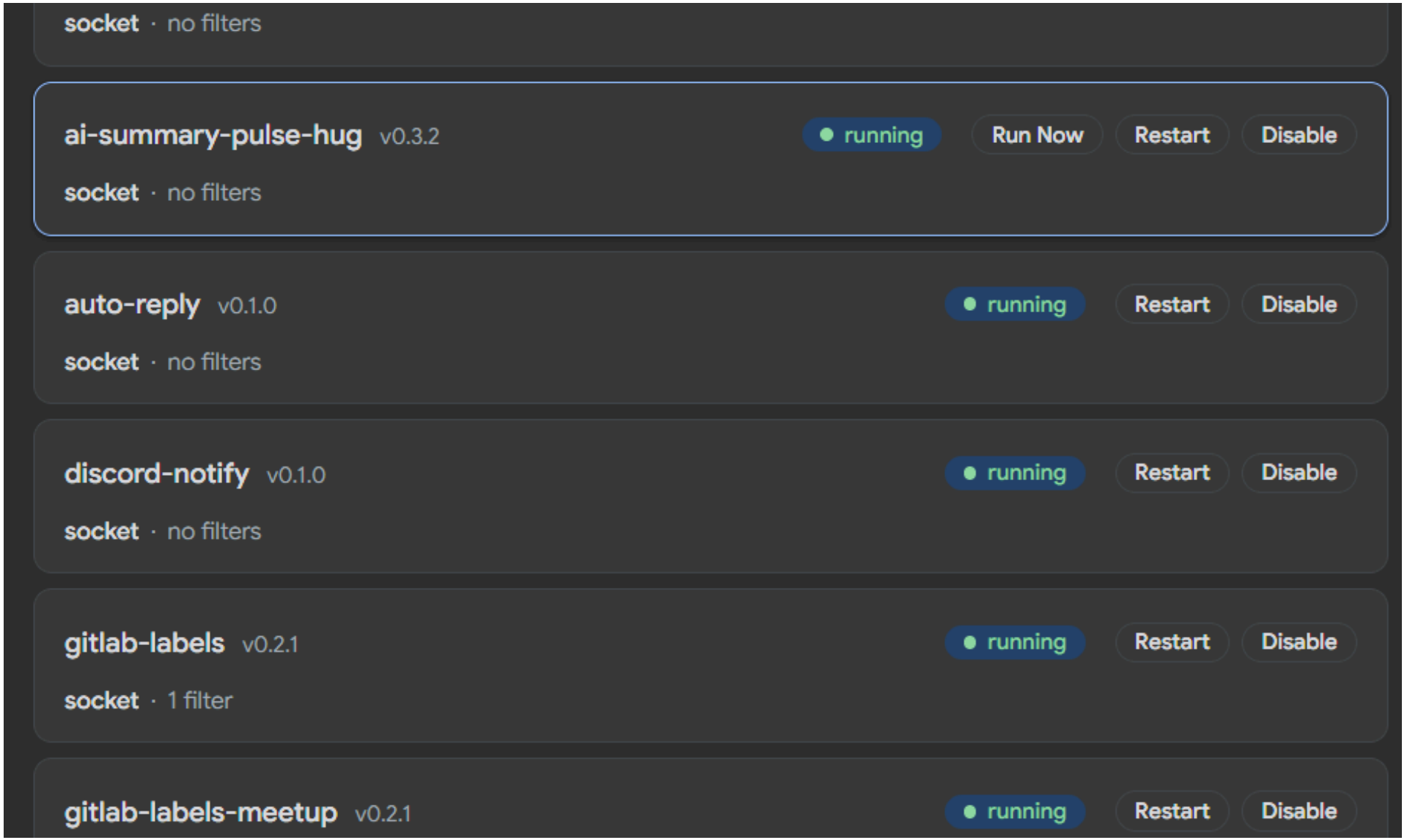


go-mail - plugins

- go-mail runs each plugin as a separate, supervised process.
- Plugins can be written in any language: the wire protocol is newline-delimited JSON (NDJSON), one `Envelope` object per line.

- The host spawns the process (or dials an already-running one), performs
  - init` handshake to learn the plugin's manifest,
  - calls `evaluate` for each email,
  - may send `batch_complete` / `scheduled_tick` events
  - finally `shutdown`.
  - Plugins may call back into the host through `host.*` RPC.

go-mail - plugins



Transports

Selected per plugin via `transport:` in `plugins.yml`.

transport	behaviour
`socket` (default)	The host allocates a Unix socket, exports its path in `GO_MAIL_PLUGIN_SOCKET`, spawns `command`, then dials it.
`stdio`	NDJSON over the child's stdin/stdout; the child's stderr is forwarded to the host log.
external	No `command:`; set `address: unix:///path` to dial an already-running plugin. TCP is not implemented.

go-mail - plugins

Envelope

Every message is one JSON object: (single line)

```
{
  "id":"...",
  "type":"request",
  "method":"evaluate",
  "payload":{...}
}
```

field	meaning
`id`	Correlation id.
`type`	`request`, `response`, or `event`.
`method`	`init`, `evaluate`, `shutdown`, `host.<name>`, `batch_complete`, `scheduled_tick`, …
`payload`	Method-specific JSON object; omitted when empty.

Handshake

Once per connection, the host sends:

```
{"id":"1","type":"request","method":"init","payload":{"host_version":"...","config":{}}}
```

The plugin must answer with its manifest:

```
{"id":"1","type":"response","method":"init","payload":{"name":"...","version":"..."}}
```

`config` is the operator's `plugins.yml` `config:` block re-encoded as JSON.

go-mail - plugins

Manifest

field	meaning
`name`	Plugin name
`version`	Plugin version; part of the verdict-cache key.
`actions`	Action types this plugin may return.
`wants`	Per-email fields : any of `envelope`, `headers`, `body`.
`cacheable`	plugin, version, account, mailbox, uid, filter_config.
`run_last`	Defers filters referencing this plugin
`rerunnable`	Disables the host's run-once-per-email guard.
`rescan_interval`	re-evaluate this plugin's filters
`wants_batch_complete`	Receive a `batch_complete` event
`wants_mailbox_state`	mailbox snapshot (honours `wants`).
`host_methods`	`host.*` methods this plugin may call
`wants_schedule`	Receive `scheduled_tick` events.
`schedule`	Tick cadence: a Go duration (`10m`) or cron (`@daily`, `0 8 * * 1-5`).

Evaluate

- host asks the plugin to judge one email:
- plugin answers:

field	meaning
`matched`	Whether the `plugin:<name>` condition matches
`actions`	Actions to apply when matched
`reason`	Human-readable explanation, shown as the condition's value in logs and the UI
`error`	Non-empty means the evaluation failed; the host logs it as an error

An empty payload (`{}`) means "no match, no actions"

Evaluate

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Thank you!

Questions & Answers

